ECON 2050 – Mathematical Economics

Optimization IV: Nonlinear Programming, Endgame

A (particularly relevant) special class of problems

In economics, finance, and business, we commonly work with nonnegative variables such as quantities and prices. Therefore, we often have to solve problems of the form

$$\max_{\boldsymbol{x} \in S} f(\boldsymbol{x}) \qquad \text{s.t.} \quad g_1(\boldsymbol{x}) \leq 0, \ldots, g_m(\boldsymbol{x}) \leq 0 \tag{$\diamondsuit$}$$

$$x_1 \geq 0, \ldots, x_n \geq 0 \tag{1}$$

where $S \subseteq \mathbb{R}^n$ is open and $f, g_1, \ldots, g_m : S \to \mathbb{R}$ are continuously differentiable.

Problems of this form are so ubiquitous that, although we can tackle them with our previous theorem on inequality constraints, a special theorem has been derived for them.

To start with, consider a constrained maximization problem

$$\max_{(x_1,x_2)\in\mathbb{R}^2} f(x_1,x_2) \qquad \text{subject to} \quad g(x_1,x_2)\leq 0,\ x_1\geq 0,\ x_2\geq 0.$$

We could let $n_1(x_1,x_2)=-x_1$ and $n_2(x_1,x_2)=-x_2$, and the problem becomes

$$\max_{(x_1,x_2)\in\mathbb{R}^2} f(x_1,x_2) \qquad \text{subject to} \quad g(x_1,x_2)\leq 0,\ n_1(x_1,x_2)\leq 0,\ n_2(x_1,x_2)\leq 0.$$

The Lagrange conditions for this problem are

$$\begin{gathered}
\frac{\partial f(x_1,x_2)}{\partial x_1}-\lambda\frac{\partial g(x_1,x_2)}{\partial x_1}+\nu_1=0\\
\frac{\partial f(x_1,x_2)}{\partial x_2}-\lambda\frac{\partial g(x_1,x_2)}{\partial x_2}+\nu_2=0\\
g(x_1,x_2)\leq 0,\ \lambda\geq 0,\ x_1\geq 0,\ x_2\geq 0\\
\lambda g(x_1,x_2)=0,\ \nu_1 x_1=0,\ \nu_2 x_2=0
\end{gathered}$$

The first two equalities can be written as

$$\frac{\partial f(x_1,x_2)}{\partial x_1} - \lambda \frac{\partial g(x_1,x_2)}{\partial x_1} = -\nu_1, \quad \frac{\partial f(x_1,x_2)}{\partial x_2} - \lambda \frac{\partial g(x_1,x_2)}{\partial x_2} = -\nu_2 \qquad (\triangle)$$

Since $\nu_1, \nu_2 \geq 0$, $(\triangle)$ is equivalent to

$$\begin{aligned} \frac{\partial f(x_1,x_2)}{\partial x_1} - \lambda \frac{\partial g(x_1,x_2)}{\partial x_1} &\leq 0 \\ \frac{\partial f(x_1,x_2)}{\partial x_2} - \lambda \frac{\partial g(x_1,x_2)}{\partial x_2} &\leq 0 \end{aligned} \quad \Longleftrightarrow \quad \nabla f(x_1,x_2) - \lambda \nabla g(x_1,x_2) \leq 0.$$

Again by $(\triangle)$, the two complementary slackness conditions, $\nu_1 x_1 = 0$ and $\nu_2 x_2 = 0$, become

$$x_1 \left(\frac{\partial f(x_1,x_2)}{\partial x_1} - \lambda \frac{\partial g(x_1,x_2)}{\partial x_1} \right) = 0 \quad \text{and} \quad x_2 \left(\frac{\partial f(x_1,x_2)}{\partial x_2} - \lambda \frac{\partial g(x_1,x_2)}{\partial x_2} \right) = 0$$

Bringing back the other four conditions, $g(x_1,x_2) \leq 0$, $\lambda \geq 0$, $x_1 \geq 0$, $x_2 \geq 0$, and $\lambda g(x_1,x_2) = 0$, we obtained a new set of conditions *without the Lagrange multipliers on the nonnegativity constraints, ν_1 and ν_2.*

Karush-Kuhn-Tucker (KKT) conditions

Going back to the more general maximization problem ($\diamondsuit$), we see that a set of *necessary* conditions for $\boldsymbol{x}^* \geq 0$ being a solution to the non-linear program *with non-negative variables* is that there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \geq 0$ such that

$$\nabla f(\boldsymbol{x}^*) - \lambda_1^* \nabla g_1(\boldsymbol{x}^*) - \lambda_2^* \nabla g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* \nabla g_m(\boldsymbol{x}^*) \leq 0,$$

$$x_i^* \left(\frac{\partial f(\boldsymbol{x}^*)}{\partial x_i} - \lambda_1^* \frac{\partial g_1(\boldsymbol{x}^*)}{\partial x_i} - \lambda_2^* \frac{\partial g_2(\boldsymbol{x}^*)}{\partial x_i} - \cdots - \lambda_m^* \frac{\partial g_m(\boldsymbol{x}^*)}{\partial x_i} \right) = 0 \quad \text{for all } i = 1, \ldots, n,$$

$$g_j(\boldsymbol{x}^*) \leq 0 \quad \text{for all } j = 1, \ldots, m,$$

$$\lambda_j^* g_j(\boldsymbol{x}^*) = 0 \quad \text{for all } j = 1, \ldots, m.$$

These conditions are sometimes called the **Karush-Kuhn-Tucker (KKT) conditions**.

Alternatively, one can also set up the Lagrangian *without considering the nonnegativity constraints*:

$$\tilde{\mathcal{L}}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) - \lambda_1 g_1(\boldsymbol{x}) - \lambda_2 g_2(\boldsymbol{x}) - \cdots - \lambda_m g_m(\boldsymbol{x}),$$

and the KKT conditions can be expressed in the following equivalent way: if $\boldsymbol{x}^* \geq 0$ is a solution to the non-linear program *with non-negative variables*, then there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \geq 0$ such that

$$\frac{\partial \tilde{\mathcal{L}}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial x_1} \leq 0, \ldots, \frac{\partial \tilde{\mathcal{L}}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial x_n} \leq 0, \tag{2}$$

$$x_1^* \frac{\partial \tilde{\mathcal{L}}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial x_1} = 0, \ldots, x_n^* \frac{\partial \tilde{\mathcal{L}}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial x_n} = 0, \tag{3}$$

$$\frac{\partial \tilde{\mathcal{L}}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial \lambda_1} \geq 0, \ldots, \frac{\partial \tilde{\mathcal{L}}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial \lambda_m} \geq 0, \tag{4}$$

$$\lambda_1^* \frac{\partial \tilde{\mathcal{L}}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial \lambda_1} = 0, \ldots, \lambda_m^* \frac{\partial \tilde{\mathcal{L}}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial \lambda_m} = 0. \tag{5}$$

Theorem

Let S be an open subset of $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ be continuously differentiable functions. Suppose

- *either each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is an affine function,*
- *or each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is continuously differentiable, and the set of gradients of constraint functions (including the nonnegative constraints) binding at $\boldsymbol{x}^*$ is linearly independent.*

If $\boldsymbol{x}^* \in S$ and it is a solution to problem

$$\max_{\boldsymbol{x} \in S} \; f(\boldsymbol{x}) \qquad \text{s.t. } \; g_j(\boldsymbol{x}) \leq 0 \text{ for } j = 1, \ldots, m, \text{ and } \boldsymbol{x} \geq 0.$$

Then there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \geq 0$ such that $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ satisfies the Karush-Kuhn-Tucker (KKT) conditions.

Example: Let $u(x_1, x_2)$ be the utility function of a consumer, where x_1 and x_2 are the quantity of goods 1 and 2, respectively. For each $i = 1, 2$, $p_i > 0$ is the price of good i, and $w > 0$ is the consumer's wealth or income. The consumer's consumption choice problem is

$$\begin{aligned} \max_{(x_1,x_2)\in\mathbb{R}^n} \quad & u(x_1, x_2) \\ \text{s.t.} \quad & p_1 x_1 + p_2 x_2 \leq w \\ & x_k \geq 0 \text{ for } k = 1, 2. \end{aligned}$$

and the KKT conditions are

$$\begin{aligned} &\frac{\partial u(x_1, x_2)}{\partial x_1} - \lambda p_1 \leq 0, \qquad x_1 \left(\frac{\partial u(x_1, x_2)}{\partial x_1} - \lambda p_1 \right) = 0, \qquad x_1 \geq 0, \\ &\frac{\partial u(x_1, x_2)}{\partial x_2} - \lambda p_2 \leq 0, \qquad x_2 \left(\frac{\partial u(x_1, x_2)}{\partial x_2} - \lambda p_2 \right) = 0, \qquad x_2 \geq 0, \\ &p_1 x_1 + p_2 x_2 - w \leq 0, \qquad \lambda(p_1 x_1 + p_2 x_2 - w) = 0, \qquad \lambda \geq 0. \end{aligned}$$

If we can use the extreme value theorem to conclude that a solution exists, similar to the previous lecture, we have a useful "cookbook" procedure for solving non-linear programs with non-negative variables:

We first check if *all* constraint functions are affine. If so,

- Write down the KKT conditions.
- Every $\boldsymbol{x}$ that (together with some $\boldsymbol{\lambda}$) satisfies the KKT conditions is a candidate solution.
- The candidates at which the value of the objective function are the largest are the solutions.

If not:

- Check whether the set of gradients of the constraint functions is linearly independent; identify the points at which it is not (note that the non-negativity constraints $\boldsymbol{x} \geq 0$ also need to be considered).
- Write down the KKT conditions.
- Every $\boldsymbol{x}$ that violates the linear independence condition or that (together with some $\boldsymbol{\lambda}$) satisfies the KKT conditions is a candidate solution.
- The candidates at which the value of the objective function are the largest are the solutions.

Example: Consider the maximization problem

$$\begin{aligned} \max_{(x_1,x_2)\in\mathbb{R}^2} \quad & x_1x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 6 \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

- Observe that the constraint functions are affine, so the KKT conditions are necessary; that is, every solution of the maximization problem must satisfy the KKT conditions.
- The KKT conditions are

$$\begin{aligned} & x_2 - \lambda \leq 0, \qquad x_1(x_2 - \lambda) = 0, \qquad x_1 \geq 0, \\ & x_1 - \lambda \leq 0, \qquad x_2(x_1 - \lambda) = 0, \qquad x_2 \geq 0, \\ & x_1 + x_2 - 6 \leq 0, \qquad \lambda(x_1 + x_2 - 6) = 0, \qquad \lambda \geq 0. \end{aligned}$$

- If $x_1 > 0$, since $x_1(x_2 - \lambda) = 0$, we must have $x_2 = \lambda$. If $x_2 = \lambda = 0$, then $x_1 - \lambda \leq 0$ implies that $x_1 \leq 0$, a contradiction. Thus, $x_2 > 0$.
- But then $x_2(x_1 - \lambda) = 0$ implies that $x_1 = \lambda = x_2 > 0$.

- Because $\lambda > 0$, by complementary slackness, $6 - x_1 - x_2 = 0$. Therefore, $(x_1^*, x_2^*, \lambda^*) = (3, 3, 3)$ satisfies the KKT conditions.
- Now suppose instead $x_1 = 0$. If $x_2 > 0$, by complementary slackness, $x_2(x_1 - \lambda) = 0$ implies $\lambda = x_1 = 0$.
- But then since $x_2 - \lambda \leq 0$, it must be that $x_2 \leq 0$, a contradiction. Thus, $x_2^* = \lambda^* = 0$.
- Therefore, $(x_1^*, x_2^*, \lambda^*) = (0, 0, 0)$ also satisfies the KKT conditions.
- Because the objective function is continuous and the constraint set is compact (draw a picture), by the extreme value theorem a solution exists.
- Then since the KKT conditions are necessary, all solutions must be among the points at which the KKT conditions are satisfied.
- Because the value of the objective function at $(x_1, x_2) = (3, 3)$ is strictly greater than that at (0, 0), the unique solution of the problem is $(x_1^*, x_2^*) = (3, 3)$.

Exercise: Consider the maximization problem

$$\max f(x,y) = x + y \qquad \text{s.t.} \quad x^2 + y^2 \leq 2, x \geq 0,\ y \geq 0$$

- Denote the three constraint functions by $g(x,y) = x^2 + y^2 - 2$, $n_1(x,y) = -x$, $n_2(x,y) = -y$.
- The linear independence condition is satisfied at all relevant points:
  - If exactly 1 constraint binds, the corresponding gradient is non-zero:
    - ⋆ If $x = 0$ is the only binding constraint, since $\nabla n_1(x,y) = (-1,0)$, the set of gradients of binding constraint functions is $\{(-1,0)\}$, which is linearly independent.
    - ⋆ If $y = 0$ is the only binding constraint, since $\nabla n_2(x,y) = (0,-1)$, the set of gradients of binding constraint functions is $\{(0,-1)\}$, which is linearly independent.
    - ⋆ If $x^2 + y^2 = 2$ is the binding constraint, since $\nabla g(x,y) = (2x, 2y)$, it is only zero when $(x,y) = (0,0)$, but the constraint doesn't bind at this point. Thus, $\{\nabla g(x,y)\}$ is linearly independent when $g(x,y) \leq 0$ is binding.

- The linear independence condition is satisfied at all relevant points:
  - If exactly 2 constraints bind, then the set of gradients of binding constraint functions is always linearly independent:
    - ★ If $x = 0$ and $y = 0$, since $\nabla n_1(x, y) = (-1, 0)$ and $\nabla n_2(x, y) = (0, -1)$ the set of gradients of binding constraint functions is $\{(-1, 0), (0, -1)\}$, which is linearly independent
    - ★ If $x = 0$ and $x^2 + y^2 = 2$ then $y = \sqrt{2}$ and since $\nabla g(x, y) = (2x, 2y)$ and $\nabla n_1(x, y) = (-1, 0)$,

      the set of gradients of constraint functions binding at $(x, y) = (0, \sqrt{2})$ is

      $$\{\nabla g(0, \sqrt{2}), \nabla n_1(0, \sqrt{2})\} = \{(0, 2\sqrt{2}), (-1, 0)\},$$

      which is linearly independent.
    - ★ If $y = 0$ and $x^2 + y^2 = 2$ then $x = \sqrt{2}$ $\nabla g(x, y) = (2x, 2y)$ and $\nabla n_2(x, y) = (0, -1)$,

      the set of gradients of constraint functions binding at $(x, y) = (\sqrt{2}, 0)$ is

      $$\{\nabla g(\sqrt{2}, 0), \nabla n_2(\sqrt{2}, 0)\} = \{(2\sqrt{2}, 0), (0, -1)\},$$

      which is linearly independent.
  - It is not possible that all 3 constraints bind: $x = y = 0$ implies $x^2 + y^2 < 2$.

- KKT conditions: $x \geq 0$, $y \geq 0$, $\lambda \geq 0$,

$$1 - 2\lambda x \leq 0 \tag{6}$$
$$x(1 - 2\lambda x) = 0 \tag{7}$$
$$1 - 2\lambda y \leq 0 \tag{8}$$
$$y(1 - 2\lambda y) = 0 \tag{9}$$
$$x^2 + y^2 \leq 2, \tag{10}$$
$$\lambda(x^2 + y^2 - 2) = 0 \tag{11}$$

- From (6) and (8) we infer that $\lambda > 0$, $x > 0$, $y > 0$. Therefore,

$$1 - 2\lambda x = 0, \qquad 1 - 2\lambda y = 0, \qquad x^2 + y^2 = 2.$$

  We obtain the unique point that satisfies KKT: $(x^*, y^*, \lambda^*) = (1, 1, \frac{1}{2})$.
- Since the constraint set is compact and the objective function is continuous, $(1, 1)$ is the unique solution (extreme value theorem).

Sufficient conditions

In lecture 8, we saw that when the objective function is concave, the "first-order conditions" become sufficient; i.e., every critical point is an optimal solution.

Since the Lagrange/KKT conditions admit a similar interpretation to the first-order conditions, it's natural to expect that a similar result should hold here.

Indeed, under further assumptions on the constraint functions, we could obtain such "sufficiency" results.

- The rough idea of these conditions is that they guarantee that the constraint set is convex, implying that the concave objective function is well defined, and that local optimality implies global optimality.

Theorem

Let S be an open subset of $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ be a continuously differentiable concave function, and each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is a continuously differentiable convex function. If $\boldsymbol{x}^* \in S$ and there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \geq 0$ such that

$$\begin{gathered}\nabla f(\boldsymbol{x}^*) - \lambda_1^* \nabla g_1(\boldsymbol{x}^*) - \lambda_2^* \nabla g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* \nabla g_m(\boldsymbol{x}^*) = 0, \\ g_i(\boldsymbol{x}^*) \leq 0 \quad \text{for all } i = 1, \ldots, m, \\ \lambda_i^* g_i(\boldsymbol{x}^*) = 0 \quad \text{for all } i = 1, \ldots, m,\end{gathered}$$

then $\boldsymbol{x}^*$ is a solution to problem

$$\max_{\boldsymbol{x} \in S} \; f(\boldsymbol{x}) \qquad \text{s.t. } \; g_i(\boldsymbol{x}) \leq 0 \text{ for } i = 1, \ldots, m.$$

Proof: Denote the constraint set by $C \equiv \{\boldsymbol{x} \in S : g_i(\boldsymbol{x}) \leq 0 \text{ for } i = 1, \ldots, m\}$.
Recall the alternative interpretation of the first equality in the Lagrange conditions: it is the first-order condition of the Lagrangian

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}^*) = f(\boldsymbol{x}) - \lambda_1^* g_1(\boldsymbol{x}) - \lambda_2^* g_2(\boldsymbol{x}) - \cdots - \lambda_m^* g_m(\boldsymbol{x}).$$

Note that we are taking $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*)$ as constants here so $\mathcal{L}$ is only a function of $\boldsymbol{x}$.

Because for each $i = 1, \ldots, m$, g_i is convex, $-g_i$ is concave. Therefore, $\mathcal{L}$, as the sum of concave functions, is also concave. Consequently, since $\boldsymbol{x}^*$ is a critical point of $\mathcal{L}$, it maximizes $\mathcal{L}$ on S.

Since $C \subseteq S$, $\boldsymbol{x}^*$ also maximizes $\mathcal{L}$ on C, meaning that $\mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) \geq \mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}^*)$ for all $\boldsymbol{x} \in C$. Then

$$f(\boldsymbol{x}^*) = \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) \geq \mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}^*) \geq f(\boldsymbol{x})$$

for all $\boldsymbol{x} \in C$, where the equality follows from complementary slackness (which implies that $\lambda_1^* g_1(\boldsymbol{x}^*) - \lambda_2^* g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* g_m(\boldsymbol{x}^*) = 0$), and the second inequality holds because $\lambda_i^* \geq 0$ and $g_i(\boldsymbol{x}) \leq 0$ for each $i = 1, \ldots, m$. Thus, $\boldsymbol{x}^*$ is a solution to the maximization problem.

An analogous result can be obtained for points satisfying the KKT conditions.

Theorem

Let S be an open subset of $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ be a continuously differentiable concave function, and each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is a continuously differentiable convex function. If $\boldsymbol{x}^* \in S$ and there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \geq 0$ such that

$$\nabla f(\boldsymbol{x}^*) - \lambda_1^* \nabla g_1(\boldsymbol{x}^*) - \lambda_2^* \nabla g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* \nabla g_m(\boldsymbol{x}^*) \leq 0,$$

$$x_i^* \left(\frac{\partial g_1(\boldsymbol{x}^*)}{\partial x_i} - \lambda_2^* \frac{\partial g_2(\boldsymbol{x}^*)}{\partial x_i} - \cdots - \lambda_m^* \frac{\partial g_m(\boldsymbol{x}^*)}{\partial x_i} \right) = 0 \quad \text{for all } i = 1, \ldots, n,$$

$$g_j(\boldsymbol{x}^*) \leq 0 \quad \text{for all } j = 1, \ldots, m,$$

$$\lambda_j^* g_j(\boldsymbol{x}^*) = 0 \quad \text{for all } j = 1, \ldots, m,$$

then $\boldsymbol{x}^*$ is a solution to problem

$$\max_{\boldsymbol{x} \in S} f(\boldsymbol{x}) \qquad \text{s.t. } g_i(\boldsymbol{x}) \leq 0 \text{ for } i = 1, \ldots, m, \text{ and } \boldsymbol{x} \geq 0.$$

Example: We saw from the exercise on slide 12 that for the maximization problem

$$\max f(x,y) = x + y \qquad \text{s.t.} \quad x^2 + y^2 \leq 2,\ x \geq 0,\ y \geq 0,$$

the only point that satisfies the KKT conditions is $(x^*, y^*) = (1, 1)$.

To conclude that $(1, 1)$ is indeed a solution, we can

- either apply the extreme value theorem like what we did in the exercise,
- or note that the objective function is concave and the constraint function $g(x,y) = x^2 + y^2 - 2$ is convex, and then apply the theorem on the previous slide.

With equality constraints, we need to further assume that the constraint functions are affine.

- Otherwise, the constraint set need not be convex.

Theorem

Let S be an open subset of $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ be a continuously differentiable concave function, and each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is an affine function.

If $\boldsymbol{x}^* \in S$ and there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \in \mathbb{R}^m$ such that $g_i(\boldsymbol{x}^*) = 0$ for each $i = 1, \ldots, m$, and

$$\nabla f(\boldsymbol{x}^*) - \lambda_1 \nabla g_1(\boldsymbol{x}^*) - \lambda_2 \nabla g_2(\boldsymbol{x}^*) - \cdots - \lambda_m \nabla g_m(\boldsymbol{x}^*) = 0,$$

then $\boldsymbol{x}^*$ is a solution to problem

$$\max_{\boldsymbol{x} \in S} f(\boldsymbol{x}) \qquad \text{s.t. } g_i(\boldsymbol{x}) = 0 \text{ for } i = 1, \ldots, m.$$

Example: Let's revisit a maximization problem we saw in the previous lecture:

$$\max_{(x_1,x_2)\in\mathbb{R}^2} f(x_1,x_2) = -2x_1^2 - 3x_2^2 \qquad \text{s.t.} \quad g(x_1,x_2) = x_1 + x_2 - 1 = 0.$$

- Since the constraint functions are affine, we know that every solution satisfies the Lagrange conditions.
- Because the objective function f is concave and the constraint function g is affine, every point satisfying the Lagrange conditions is a solution.
- In the previous lecture, we identified the unique point satisfying the Lagrange conditions: $(x_1^*, x_2^*, \lambda^*) = (3/5, 2/5, 12/5)$.
- Thus, the unique solution is $(x_1^*, x_2^*) = (3/5, 2/5)$.

The example on the previous slide reveals a much more general feature:

Corollary

Let S be an open subset of $\mathbb{R}^n$, and suppose $f : S \to \mathbb{R}$ be a continuously differentiable concave function, and each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is an affine function. Then

- *$\boldsymbol{x}^*$ is a solution to the constrained optimization problem with equality constraints if and only if there exists $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ such that $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ satisfies the corresponding Lagrange conditions;*
- *$\boldsymbol{x}^*$ is a solution to the constrained optimization problem with inequality constraints if and only if there exists $\boldsymbol{\lambda}^* \geq 0$ such that $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ satisfies the corresponding Lagrange conditions;*
- *$\boldsymbol{x}^*$ is a solution to the constrained optimization problem with inequality constraints and nonnegativity constraints if and only if there exists $\boldsymbol{\lambda}^* \geq 0$ such that $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ satisfies the corresponding KKT conditions.*

That is, the set of points satisfy the Lagrange/KKT conditions coincide with the set of solutions of the constrained optimization problem.

Consider a linear programming problem $\max\{\boldsymbol{c} \cdot \boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}, \boldsymbol{x} \geq 0\}$. To be more concrete, let

$$\boldsymbol{c} = (c_1, c_2), \quad \boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad \text{and} \quad \boldsymbol{b} = (b_1, b_2);$$

that is, the problem can be written as

$$\begin{aligned} \max_{(x_1,x_2)\in\mathbb{R}^2} \quad & c_1 x_1 + c_2 x_2 \\ \text{s.t.} \quad & a_{11} x_1 + a_{12} x_2 \leq b_1 \\ & a_{21} x_1 + a_{22} x_2 \leq b_2 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

Its dual is $\min\{\boldsymbol{y} \cdot \boldsymbol{b} : \boldsymbol{y}\boldsymbol{A} \geq \boldsymbol{c}, \boldsymbol{y} \geq 0\}$, which can be written as

$$\begin{aligned} \min_{(y_1,y_2)\in\mathbb{R}^2} \quad & b_1 y_1 + b_2 y_2 \\ \text{s.t.} \quad & a_{11} y_1 + a_{21} y_2 \geq c_1 \\ & a_{12} y_1 + a_{22} y_2 \geq c_2 \\ & y_1 \geq 0, y_2 \geq 0 \end{aligned}$$

KKT for linear programming problems

By the result on slide 22, since the objective function is linear and therefore concave, the solutions of this problem are precisely the points satisfying the KKT conditions.

The KKT conditions are

$$\begin{aligned}
c_1 - \lambda_1 a_{11} - \lambda_2 a_{21} \leq 0, \quad & x_1(c_1 - \lambda_1 a_{11} - \lambda_2 a_{21}) = 0, \quad & x_1 \geq 0; \\
c_2 - \lambda_1 a_{12} - \lambda_2 a_{22} \leq 0, \quad & x_2(c_2 - \lambda_1 a_{12} - \lambda_2 a_{22}) = 0, \quad & x_2 \geq 0; \\
a_{11}x_1 + a_{12}x_2 - b_1 \leq 0, \quad & \lambda_1(a_{11}x_1 + a_{12}x_2 - b_1) = 0, \quad & \lambda_1 \geq 0; \\
a_{21}x_1 + a_{22}x_2 - b_2 \leq 0, \quad & \lambda_2(a_{21}x_1 + a_{22}x_2 - b_2) = 0, \quad & \lambda_2 \geq 0.
\end{aligned}$$

Observe that (by changing λ_1, λ_2 to y_1, y_2) the KKT conditions are precisely the dual constraints, primal constraints, and the complementary slackness conditions.

This exercise further elucidates the essence of the Lagrange/KKT approach:

- We write down all the primal constraints, dual constraints, and complementary slackness conditions; these are called Lagrange/KKT conditions.
- We know from our previous discussion that these must be satisfied by any pair of primal solution and dual solution, $(\boldsymbol{x}^*, \boldsymbol{y}^*)$ (when the constraint functions are not affine, further technical assumptions needed to guarantee that the dual constraints make sense).
- Therefore, by checking which points satisfy the Lagrange/KKT conditions, we rule out most points in the constraint set; that is, we use these conditions to "screen" irrelevant points.
- Then by comparing the objective function's values at (usually small amount of) "surviving points," we obtain the solutions.
- In particular, if the objective function is concave and the constraint functions are affine, solving Lagrange/KKT conditions directly yield the solutions of the maximization problem.